

Milestone 2: Final Memo

BUS 445 D100: Customer Analytics

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July 27, 2026

YouTube video link: <https://www.youtube.com/watch?v=6lwNbpzUri4>

MEMORANDUM

To: Karen Doyle, VP Customer Base Management
Raj Patel, Director of Retention Operations
Sandra Liu, Chief Financial Officer

From: Aurora Advisory Consulting Team 5

Date: July 27, 2026

Subject: Reducing High-Value Customer Churn Through Targeted Retention and Network Infrastructure Investment

We recommend that Cell2Cell take a forward-thinking approach to retaining its most valuable customers, especially those showing signs of churn. Immediately, the company should offer targeted special offers to high-value customers before a threat of leaving while also investing in sensible network infrastructure improvements in the areas where service problems are most common. They should start randomized testing experiments to develop broader, scalable retention and special offers that produce the highest long-term customer value. Over the next three years, this strategy could preserve about **\$3.1 million** in revenue.

High-CLV customers with low retention are heavy usage users and experience more dropped calls, blocked calls, unanswered calls and call-waiting events than similar high-CLV customers with high retention. These customers *appear* to be getting a worse service experience, which may contribute to their higher churn risk. Many can be kept with an appropriate offer by Cell2Cell's call team, thus the importance of developing more standardized retention offers.

Although Cell2Cell should use the immediate retention and special offer plan to reduce short-term churn and protect revenue, the bigger issue is infrastructure. As these network failures and overload are highly correlated to churn, and if the company does not upgrade its network infrastructure to accommodate even modest growth, today's high-retention customers could become tomorrow's high-risk customers. Cell2Cell needs to invest in network capacity and reliability, before service problems become widespread and start eroding the loyalty of its most valuable users. By waiting for network congestion to worsen with even a modest growth rate, it risks allowing service

interruptions to spread beyond today's highest-usage customers, degrading the experience of its high-retention customer base as overall network demand continues to grow.

The strategy below focuses on an immediate strategy to reduce network pressure with modest infrastructure upgrades, while keeping costs manageable and validating our hypothesis before moving to larger-scale plans.

Problem and question

Cell2Cell operates in a saturated U.S. wireless market where acquiring new customers has become slow and expensive. The company's profitability now depends far more on keeping the customers it already has than on winning new ones. The current retention approach is reactive: the company only intervenes once a customer is already trying to cancel, which is costly and does little to protect long term value. The real challenge is that churn is not equally expensive across the customer base. Losing a low value customer barely moves the needle, while losing a high value customer erases a disproportionate share of monthly revenue. The business problem is therefore not simply to reduce churn, but to identify and keep the specific high value customers who are at risk of leaving, before they leave, without spending money on customers who were never going to churn.

This leads to the single analytical question our analysis is built to answer: **among Cell2Cell's highest value customers, which ones are most at risk of churning, and what measurable characteristics separate them from comparable high value customers who stay?** Answering this requires three connected modeling steps. We first estimate each customer's probability of churning, then combine that with a Customer Lifetime Value score to find the customers who are both worth the most and least likely to stay and finally compare that at risk group against otherwise similar loyal customers to see what sets them apart across usage, network, spending, and account behavior.

We chose this question over the more common alternative of simply predicting who will churn across the whole customer base. Predicting churn alone tells the company who is at risk but not why, and not whether those customers are worth saving, so it cannot guide where to spend limited retention and network dollars. It also treats a low value churner and a high value churner as equally important, which they are not. By anchoring the question in Customer Lifetime Value and then asking what distinguishes the valuable leavers, we get something the company can act on: a ranked, explainable target list that points retention outreach and network investment at the customers whose

loss would cost the most. That is the shift from reactive damage control to a proactive, evidence-based retention strategy.

Data and approach

The Cell2Cell customer dataset, which contains **one month of customer records** (51,047) and **58 original variables**. The dependent variable was **Churn**, recorded as “Yes” or “No.”

The analysis was completed in three connected stages: predicting **customer retention probability**, estimating **customer lifetime value (CLV)**, and **segmenting high-value customers** based on Retention Rate for contrasting comparison and clearer look at who we need to target first.

CustomerID, was removed because it uniquely identifies individual records and does not help with our classification models. The remaining variables were divided into numeric and categorical features.

The original dataset was imbalanced, containing **36,336 non-churners and 14,711 churners**. A 75/25 train-validation split was used and only the training set was balanced resulting in a training sample of **11,033 churners and 11,033 non-churners**.

We fit an **XGBoost** classification model. After Hypertuning, XGBoost achieved an **AUC of 0.675**. The model produced a continuous probability of churn **Churn_Proba** where we could then add **Retention_Proba** for each customer. A Customer Lifetime Value score **CLV** for all customers was added, using:

$$\text{CLV} = 12 * (\text{ContributionMargin} * (\text{Retention_Proba} / 1 + (i = 0.0023) - \text{Retention_Proba}))$$

Where: $i = 0.023$ is annual 2.8 inflation / 12 months

And: **ContributionMargin** = **MonthlyRevenue** – **EstimatedCost** is composed of (**CustomerCareCalls*4.00** + **MonthlyMinutes*0.01**)

We filtered this active dataset to find the **High-CLV customers** (top 30 percent of highest CLV(Churn=No) customers) to focus on. The top 30 percent, which is 10,901 / (51,047 Churned = Yes or No) or 10,901 / (36,336 Churned = No) and **represents almost 30 percent of the month's total revenue**.

The final stage focused on identifying which active, **high-CLV customers were at the greatest risk of churn**. High-CLV customers were divided using a **65% predicted retention threshold**. To determine this threshold, the validation dataset was iterated

over a range of candidate **probability_thresholds** using a candidate threshold variable. At each threshold, predicted probabilities were converted into **Churn = Yes/No** classifications and compared against the true outcomes, $G(X)$. The threshold that produced the highest prediction score was selected. Although a threshold of 0.50 is commonly used by default, it resulted in poorer classification performance for this model (see Assumptions). The optimal threshold was found to be **65%**, meaning customers with predicted retention probabilities below 65% were classified as the **High CLV–Low Retention** group, while those with retention probabilities of 65% or greater formed the **High CLV–High Retention** comparison group. We then compared the two groups using:

- Number of customers in each segment
- Total monthly revenue generated by each segment
- Total company monthly revenue
- Revenue generated by the High-CLV – Low-RR segment (treated as revenue at risk)
- The percentage of total company revenue represented by the High CLV / Low RR segment

The **High CLV–High Retention** group was used as a comparison. Network and service variables: **dropped calls, blocked calls, dropped or blocked calls, unanswered calls, customer-care calls, and call-waiting calls** were compared between the groups. Normality was assessed and a **Mann-Whitney U test** was used. After significance was shown we calculated the groups means and percentage differences between groups.

Additional comparisons examined **monthly minutes, overage minutes, roaming calls, dropped calls, blocked calls, unanswered calls, customer care calls, call waiting events, dropped and blocked calls, current equipment age, handset price, handset refurbishment status, handset web capability, months in service, retention calls, retention offers accepted, monthly revenue, and total recurring charges**. Mann-Whitney U tests were performed, and group means and percentage differences were calculated to identify statistically and practically important differences between the two customer groups.

Findings

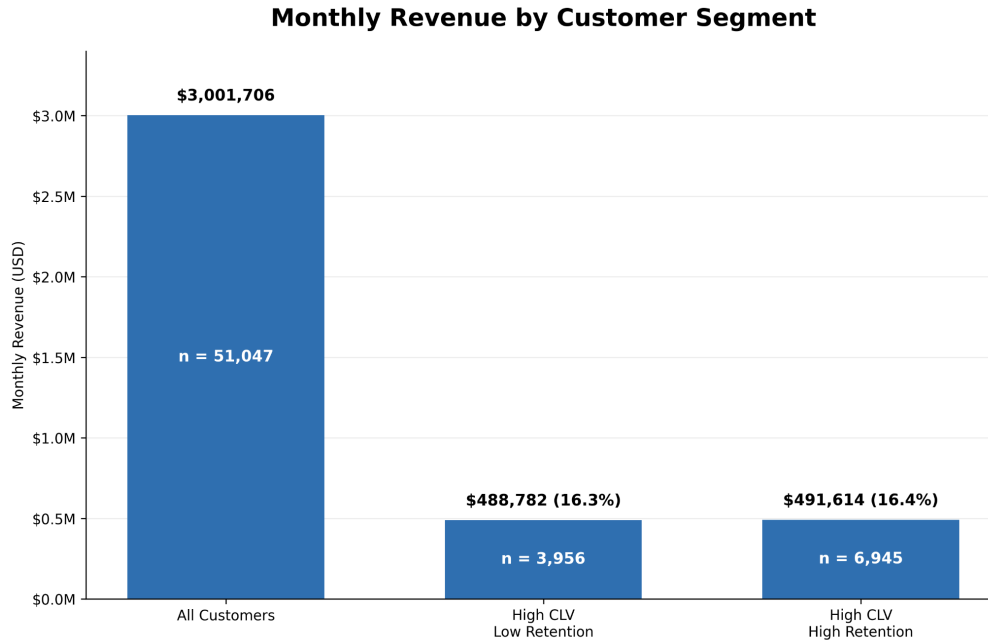
The High-CLV Segment

High-CLV customers compose a smaller portion of Cell2Cell's customer base, but they contribute a much larger share of monthly revenue. The **top 30th percentile high-CLV** customers (10,901) accounted for about **33% of monthly revenue**. Their CLV is much higher because they generate more **revenue** and **customer margin** through heavier use of **base plans**, **minute overages**, **roaming**, and other services. These customers also show more variation in how they use the network and how much they spend, useful for further analysis. **Low-CLV** customers tend to pay lower monthly fees, use fewer minutes, rarely incur extra charges, and contact customer service less often. Their behavior is more stable and less varied, which makes it harder to identify strong patterns or meaningful differences within that group. As a result, the High-CLV segment is the most important group to study, both because it accounts for more revenue and because its greater variation provides clearer evidence about customer behavior and retention risk.

Finding our Comparable Groups: Splitting the High-CLV Segment

To better understand the characteristics of customers most valuable to the business, we isolated customers with **High Customer Lifetime Value (CLV)** and divided them into two cohorts based on retention performance: **High CLV - High Retention Rate (High-RR)** and **High CLV - Low Retention Rate (Low-RR)** customers.

High CLV - Low Retention Rate (Low-RR) was the primary target because these customers combined high economic value with elevated risk. The sub-segment generated approximately **\$480,000–\$489,000 in monthly revenue**, or roughly **16% of total company revenue**, despite representing the smallest subset of the active customer population with **3,956 customers**. This paradox exists because the Revenue and Customer margin(m) is so large in this cohort it multiplies the CLV formula edging the **CLV UP**, while **Retention rate can remain DOWN**. Conversely, if the customer has a low monthly bill per month but they call Customer Service frequently this can lower their CLV.



Because both comparison groups already represent the company's most valuable customers, differences observed between them cannot be attributed to customer value alone. Their contribution to Total Monthly Revenue is almost the same. But they can provide insight into the behavioral and service characteristics associated with lower retention among otherwise highly profitable customers.

Characteristics of High CLV Customers with Lower Retention

The first major distinction between the two groups is **overall network usage**. High-CLV- Low-RR **1,040 monthly minutes (ave per customer)**, compared with **671 monthly minutes (ave per customer)** for High-CLV-High-RR, a difference of approximately **55%**. The low-retention cohort are **high-usage customers** who have higher base plans and frequently go over in minutes.

Evidence suggests network quality is strongly associated with lower retention

With this increased level of usage, the Low-RR segment experiences significantly higher rates of **every network quality metric** (mean adjusts for n_customers) examined compared with High-RR customers:

- 61% more Dropped and Blocked Calls** (18.44 vs. 11.47)
- 75% more Call Waiting Events** (4.89 vs. 2.79)

- 52% more Dropped Calls** (11.03 vs. 7.24)
- 44% more Unanswered Calls** (49.67 vs. 34.47)
- 74% more Blocked Calls** (7.26 vs. 4.16)
- 17% more Customer Care Calls** (1.74 vs. 1.48)

All differences were statistically significant according to the **Mann–Whitney U test** ($p < 0.001$). These patterns are unlikely to be explained by random variation.

These findings may not demonstrate that network issues cause customer churn. They do infer that high-value customers with lower retention have **high usage -> more network interactions -> connection related issues** than comparable customers who remain loyal. This catastrophic pattern projects into the future as a large growth in customers intensifies this issue bleeding into the High-RR group.

Additional Actionable Customer Variables

Beyond network quality metrics, several additional behavioural characteristics distinguish the Low-RR cohort.

Compared with High-RR customers, the Low-RR segment exhibits:

- 302% more roaming calls**
- 238% more overage minutes**
- 75% higher monthly revenue**
- 20% higher recurring monthly charges**
- 49% longer equipment ownership**
- 41% longer tenure with the company**

These results show that customers with lower retention are not just more active users, they represent substantially higher value accounts whose service usage extends well beyond basic monthly plans.

The Low-RR group receives **602% more retention calls** than the High-RR group in the one-month sample and accepts **436% more retention offers**, corresponding to a **72% acceptance rate**. These findings should not be interpreted as evidence that retention offers improve customer loyalty. Instead, they indicate that this segment remains highly responsive to intervention. Because the Cell2Cell case describes the current retention program as reactive, contacting customers only after they indicate an intention to leave, these results highlight an opportunity to engage high-value customers' earlier through proactive outreach.

Although the High-CLV Low-RR segment already receives substantially more retention outreach, these interventions occur only after customers have become potential churners. The Classification Model provides an opportunity to identify high-value customers earlier and deliver targeted customer offers before they reach the cancellation stage.

Projecting Retention Revenue

If we assume a Retention Rate = 0.50 for a lower floor (if no action is taken) and our target Retention Rate = 0.75 following immediate strategy implementation and the High-CLV - Low-RR's Monthly Revenue = ~488k we get our three-year projection:

Annual Retention	3-Year Discounted CLV
75.0%	\$13.245M
50.5%	\$10.145M
Difference	≈ \$3.10M

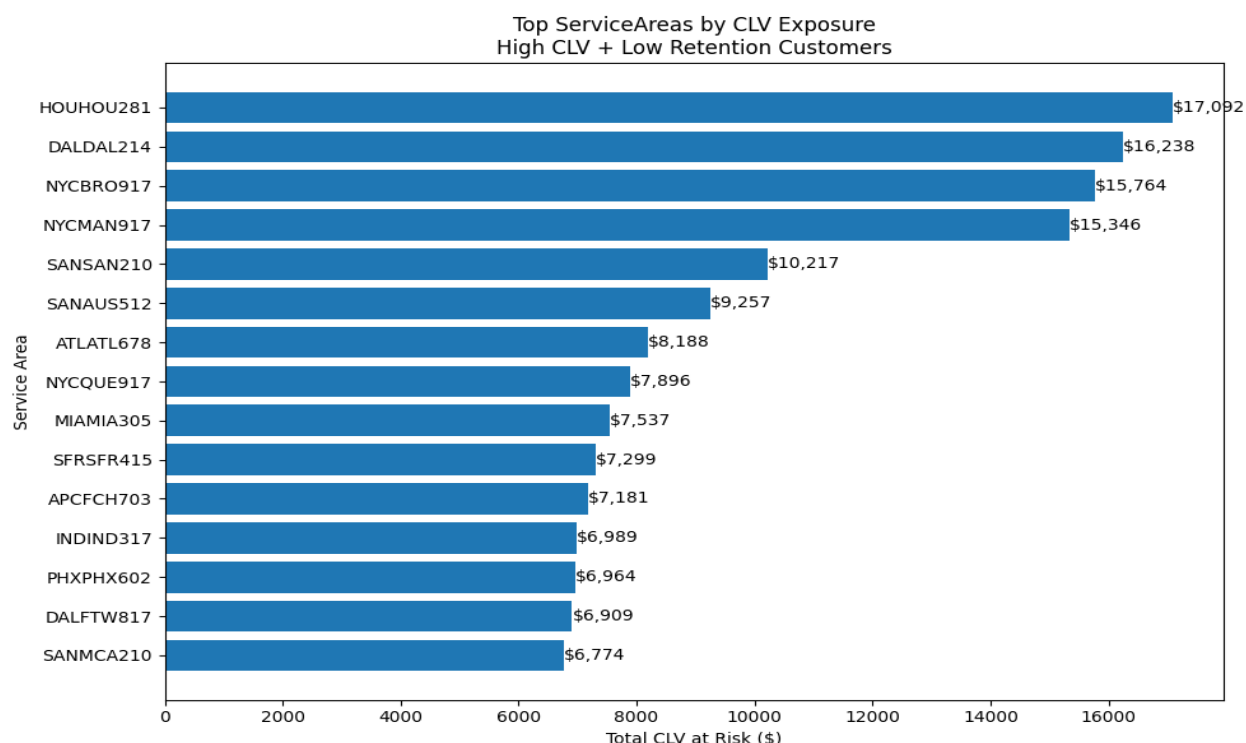
Recommendation and dollar impact

Prioritize Network Upgrades in High-Risk Service Areas

Cell2Cell should implement a phased network investment strategy targeting the 32 priority service areas identified in our analysis. These locations contain approximately 1,706 high-CLV customers with low predicted retention, representing \$7.71 million in potential lost revenue exposure over the next three years. Targeted network improvements in these areas combined with a retention strategy provide the company with an opportunity to preserve this revenue.

Rather than implementing network improvements across all 411 service areas, focusing on these ~32 high-impact zones allow Cell2Cell to direct capital and retention resources where they will have the greatest financial impact while avoiding unnecessary investment in low-density locations. Our analysis shows that network quality issues directly correlate with churn risk among high-value customers. Improvements in these

specific areas will simultaneously reduce service failures and protect revenue from Cell2Cell's most profitable segment



Small cell and network densification technologies are standard solutions for addressing capacity constraints in high-density customer areas. Industry data indicates that small cells (femtocells and picocells) range from \$5,000–\$8,000 per node, while Distributed Antenna Systems (DAS) range from \$8,000–\$15,000 per node. For a mixed urban deployment of the type Cell2Cell would undertake in its 32 priority service areas, typical costs fall within the \$7,000–\$15,000 per node range based on current vendor pricing from infrastructure providers.

Cell2Cell should partner with established telecom infrastructure consultants to conduct detailed site assessments in the 32 priority areas. External partners should manage technology selection, permitting, and deployment. Cell2Cell's role is to own the strategic decisions (which areas to prioritize, timeline), manage capital allocation, and validate performance outcomes.

Implementation Timeline and Cost

Year 1 should focus on assessment and initial deployment. Cell2Cell should contract with an external engineering firm to conduct network assessments in all 32 priority service areas during the first three months. This assessment phase will cost

approximately \$150,000–\$200,000 and identify optimal placement for small cells or DAS infrastructure. During the remainder of Year 1, Cell2Cell should deploy technology in the top 8–10 highest-density locations, representing approximately 40–60 nodes at \$300,000–\$600,000. The total Year 1 investment is approximately \$450,000–\$800,000.

Years 2 and 3 should continue the rollout to remaining priority service areas. Cell2Cell should budget approximately \$400,000–\$700,000 per year for continued deployment, bringing all priority service areas online by the end of Year 3. The cumulative three-year investment is estimated at \$1.25–\$2.2 million. The revenue at risk across these 32 areas totals \$7.71 million over three years (\$214,218 monthly, or \$2.57 million annually).

Proactively Sending Offers

While working on network improvement plans, Cell2Cell can focus on the second part of the strategy, a sustainable retention model targeted toward our target segment. To do this, a list of customers that should be prioritized will be created to optimize outreach efforts toward high revenue potential customers and those who are affected by the poor network most.

Because network improvements require time to plan and implement, proactive customer outreach can begin immediately. Rather than waiting until high-value customers initiate cancellation requests, Cell2Cell can use the Classification Model to identify customers most at risk of churn and engage them earlier with targeted interventions with “special offers”. This allows the company to reduce churn during the period in which long-term network improvements are still being completed.

These special offers can be applied to customers based on their usages. These would include plan upgrades for heavy users, device upgrade incentives for customers with older handsets, or targeted travel packages for frequent roaming customers. Customers going through network quality issues should get frequent contact and temporary retention incentives while improvement plans are ongoing. Offers should be designed to address customer needs without unnecessarily reducing Customer Lifetime Value by discounting services that customers already purchase frequently.

Optimize Proactive Customer Offers

To continuously improve retention performance, Cell2Cell should implement randomized controlled experiments (A/B testing) for customers. Rather than assuming one retention or special offer generally works, applicable customers should be randomly assigned to different incentives, such as bill credits, plan upgrades or loyalty rewards. Retention rates, offer acceptance rates, customer lifetime value, and incremental revenue should then be compared across groups to identify the most profitable retention strategy for each customer profile. Over time, these experimental results can be incorporated into the retention decision process, allowing Cell2Cell to replace generalized offers with evidence-based, segment-specific retention policies.

The success of this strategy can be monitored through the following KPIs:

- Monthly retention rate among the High CLV–Low Retention segment
- Retention offer acceptance rate
- Monthly revenue preserved
- Customer Lifetime Value retained
- Change in trends in network quality metrics such as Call Waiting Rate, Dropped Call Rate, Unanswered Call Rate, and Dropped + Blocked Call Rate within the prioritized service areas.

As network improvements decrease service interruptions, the company should expect these technical indicators to improve alongside customer retention, validating both the infrastructure investments and the targeted retention strategy. This approach also allows Cell2Cell to determine the minimum effective intervention for each customer segment. Rather than relying on costly, generalized retention offers after a customer has decided to leave, the company can identify lower-cost incentives that achieve comparable retention outcomes. Standardizing these evidence-based offers across similar customer profiles can improve the consistency of retention decisions while reducing unnecessary retention spending.

Assumptions

Cell2Cell is a top seven national provider, not a Big-three: Cell2Cell is more akin to Freedom mobile, and is not at the scale of Rogers, TELUS, Bell. Big three companies throw money at ads and sponsorships as their budget to acquire is massive.

Infrastructure expertise: This analysis assumes Cell2Cell will partner with qualified telecom engineering firms. We have not conducted detailed RF modeling or site surveys and defer those assessments to specialists.

External factors: Network improvements require permitting, vendor availability, and capital allocation. Regulatory or supply-chain delays could affect the timeline.

Retention offers effectiveness: Retention improvement percentages assume Part 2 retention strategy is executed in parallel with infrastructure upgrades. Infrastructure alone will not solve churn entirely because there is a lag on the upgrades.

Scalability: The \$7,000–\$15,000 per-node cost assumption is based on typical North American urban deployments. Rural or remote service areas may cost significantly more.

P(Response) rises when Retention Rate lowers: We covered this assumption in class. A low RR intuitively means they are looking for offers from Cell2Cell or elsewhere.

Limitations

Our technical network recommendations yield to an engineering firm: We cannot include detailed multiyear plans to upgrade the infrastructure. Our solution is based on the technical information available online.

Having a smaller one month data sample: Ideally, we would like to segment into **Low-Medium-High CLV** and then **Low-Medium-High Retention Rate** and do use Anova and Tukey's analysis within those segments.

Data age: The dataset contains records from 10+ years ago. Current network conditions, technology costs, and customer expectations may differ. Recommendations should be stress-tested against current network performance data. I.e. Canada-wide unlimited is the standard but **average minutes** can be used as proxy for **data coverage**.

AI disclosure

***Disclaimer:** AI tools were used to refine code, troubleshoot errors, and revise grammatical structure of the findings. Prompts included requests to debug code and ensure alignment of approaches with the deliverable. Approach, Final analysis and recommendations were produced and reviewed by our team.*

Resources:

https://api.ctia.org/docs/default-source/default-document-library/small-cell-deployment-regulatory-review-costs_3-12-2018.pdf
<https://www.telit.com/blog/5g-networks-guide-to-small-cell-technology/>